Vector

Scalar	Vector
① The physical quantity that has only magnitude is called scalar.	The physical quantity that has both magnitude and direction is called vector.
② They are represented by ordinary letters like A, B, --- etc.	They are represented by ordinary letters with arrow over them or by bold letters like $\vec{A}$ or **A**.
③ They are added by simple algebra rule.	They are added by vector addition rule.
④ They change due to the change in magnitude.	They changed due to the change in either magnitude or direction both.
⑤ Mass, Time, distance, Speed, Current, work, energy etc are some enamples of scalars.	Displacement, velocity, acceleration, Force, momentum, torque etc are some enamples of Vectors.

• Types of vectors

i) Null vector or Zero vector

- A vector having zero magnitude is called null vector. If $\vec{A}$ is a null vector then $|\vec{A}| = 0$

ii) Unit Vector

- A vector having unit magnitude is called unit vector. If $\vec{A}$ is the unit vector then $|\vec{A}| = 1$

The unit vector of $\vec{A}$ is given by

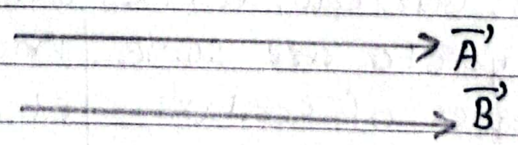
$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

iii) Collinear vectors

- The vectors lying on the same line or parallel lines are called collinear vectors. They may be parallel or antiparallel vectors.

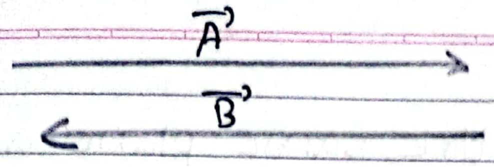
iv) Like vectors or Parallel vectors

- Vectors having same direction are called like vectors or parallel vectors.



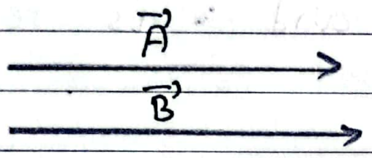
v) Unlike vectors or anti parallel vectors

- Vectors having opposite directions are called unlike or anti-parallel vectors.



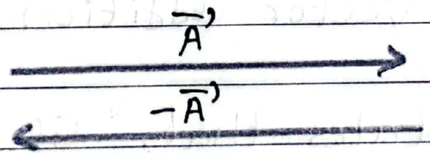
vi) Equal vectors

- Vectors are said to equal if they have same magnitude and direction. In other words, they are the like vectors having same magnitude.



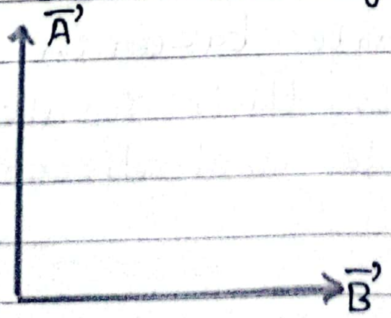
vii) Negative vector

- A vector is said to be negative vector of other vector if it has same magnitude but opposite in direction. In other words, it is the unlike ~~vectors~~ having same magnitude. The negative vector of $\vec{A}$ is represented by $-\vec{A}$.



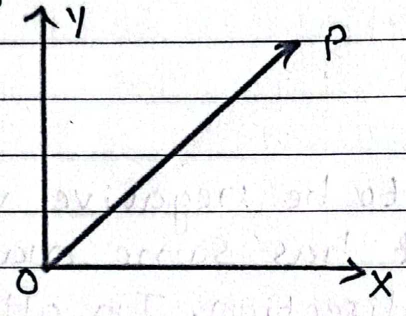
viii) Orthogonal vectors

- The vectors perpendicular to each other are called orthogonal vectors.



ix) Coplanar vectors
- Vectors lying on the same plane are called coplanar vectors.

x) Position vector
- A vector whose initial point starts from origin is called position vector. The straight line drawn from O to P gives the position vector of P and ~~*~~ is represented by $\vec{OP}$.

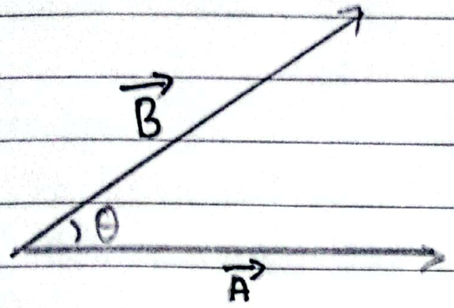


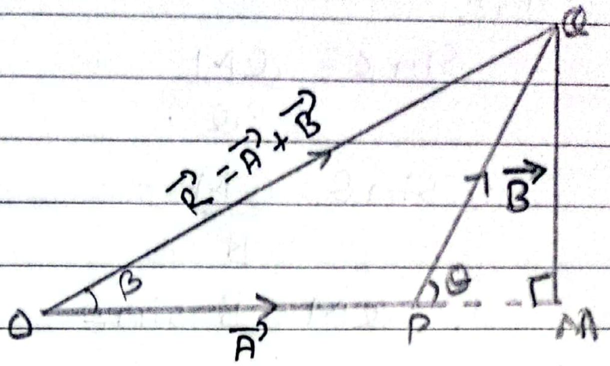
• Vector addition

① Triangle law of vector addition

Statement : It states that, "If two vectors acting at a point are represented both in magnitude and direction by the two sides of a triangle taken in same ~~direction~~ order then third side of the triangle taken in opposite order represents their resultant both in magnitude and direction.

Explanation





Consider two vectors $\vec{A}$ and $\vec{B}$ acting at a point are represented by two sides $\vec{OP}$ and $\vec{PQ}$ of a triangle OPQ. Then the third side OQ represents their resultant $\vec{R}$.

$$\therefore \vec{R} = \vec{A} + \vec{B}$$

Magnitude of $\vec{R}$

let us produce OP and draw $QM \perp OM$

In the figure,

$OP = A$

$PQ = B$

$OQ = R$

In $\triangle PQM$,

$$\cos\theta = \frac{PM}{PQ}$$

$$\text{or, } \cos\theta = \frac{PM}{B}$$

$$\therefore PM = B\cos\theta$$

And,

$$\sin\theta = \frac{QM}{PQ}$$

or, $\sin\theta = \frac{QM}{B}$

$\therefore QM = B\sin\theta$

Applying pythagorous theorem in ΔOQM,
We have,

$$OQ^2 = OM^2 + QM^2$$

or, $R^2 = (A + B\cos\theta)^2 + (B\sin\theta)^2$

or, $R^2 = A^2 + 2AB.\cos\theta + B^2\cos^2\theta + B^2\sin^2\theta$

or, $R^2 = A^2 + 2AB.\cos\theta + B^2(\cos^2\theta + \sin^2\theta)$

or, $R^2 = A^2 + 2AB.\cos\theta + B^2$

or, $R = \sqrt{A^2 + B^2 + 2AB.\cos\theta} \longrightarrow (i)$

$$\boxed{\therefore R = \sqrt{A^2 + B^2 + 2AB.\cos\theta}}$$

~~This~~ This gives magnitude of $\vec{R}$

Direction of $\vec{R}$

Let β be the angle made by $\vec{R}$ with $\vec{A}$.

In ΔOQM,

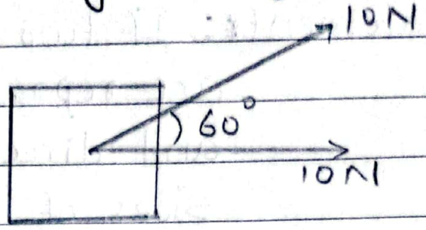
$$\tan\beta = \frac{QM}{OM}$$

or, $\tan\beta = \frac{B\sin\theta}{A + B\cos\theta}$

$\therefore \beta = \tan^{-1}\left(\frac{B\sin\theta}{A + B\cos\theta}\right) \longrightarrow (ii)$

This gives direction of $\vec{R}$

Q.n. Find the Magnitude and Direction of resultant force in the given figure.



→ Soln,

Resultant (R) = ?

Direction = ?

We know that,

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$= \sqrt{10^2 + 10^2 + 10 \times 10 \times \frac{1}{2} \times 2}$$

$$= \sqrt{100 + 100 + 100}$$

$$= \sqrt{300}$$

$$= 10\sqrt{3}$$

$$\text{Direction } (\beta) = \frac{B\sin\theta}{A + B\cos\theta}$$

$$\text{or, } \tan\beta = \frac{10 \times \sin 60}{10 + 10\cos 60}$$

$$\text{or, } \tan\beta = \frac{10 \times \frac{\sqrt{3}}{2}}{10 + 10 \times \frac{1}{2}}$$

$$\text{or, } \tan\beta = \frac{5\sqrt{3}}{15}$$

$$\text{or, } \tan\beta = \frac{\sqrt{3}}{3}$$

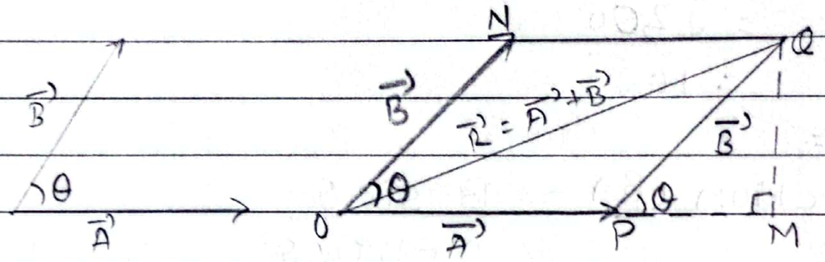
$$\text{or, } \tan\beta = \frac{1}{\sqrt{3}}$$

$$\text{or, } \beta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\therefore \beta = 30°$$

(11) Parallelogram law of vector addition

→ Statement: If two vectors acting at a point are represented both in magnitude and direction by the two adjacent sides of a parallelogram drawn through a point then diagonal of the parallelogram drawn through that point represents their resultant both in magnitude and direction.



Consider two vectors $\vec{A}$ and $\vec{B}$ inclined at an angle θ are represented by the two adjacent sides $\vec{OP}$ and $\vec{ON}$ (or $\vec{PQ}$) respectively of a parallelogram OPQN. then the diagonal $\vec{OQ}$ represents their resultant $\vec{R}$

$$\boxed{\text{i.e.- } \vec{R} = \vec{A} + \vec{B}}$$

Magnitude of $\vec{R}$

let us produce OP and draw $QM \perp PM$

In the figure,

$OP = A$

$PQ = B$

$OQ = R$

In $\triangle PQM$,

$\cos\theta = \frac{PM}{PQ}$

or, $\cos\theta = \frac{PM}{B}$

$\therefore PM = B\cos\theta$

And,

$\sin\theta = \frac{QM}{PQ}$

or, $\sin\theta = \frac{QM}{B}$

$\therefore QM = B\sin\theta$

Applying pythagoorous theorem in $\triangle OQM$

We have,

$(OQ)^2 = OM^2 + QM^2$

or, $R^2 = (A + B\cos\theta)^2 + (B\sin\theta)^2$

or, $R^2 = A^2 + 2AB\cos\theta + B^2\cos^2\theta + B^2\sin^2\theta$

or, $R^2 = A^2 + 2AB\cos\theta + B^2(\cos^2\theta + \sin^2\theta)$

or, $R^2 = A^2 + 2AB\cos\theta + B^2$

or, $R = \sqrt{A^2 + B^2 + 2AB.\cos\theta}$

$\therefore R = \sqrt{A^2 + B^2 + 2AB.\cos\theta}$ #.

This gives magnitude of $\vec{R}$

Direction of $\vec{R}$

let β be the angle made by $\vec{R}$ with $\vec{A}$.

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In ΔOQM,

$$\tan\beta = \frac{QM}{OM}$$

or, $$\tan\beta = \frac{B\sin\theta}{A + B\cos\theta}$$

$$\therefore \beta = \tan^{-1}\left(\frac{B\sin\theta}{A + B\cos\theta}\right)$$

This gives direction of $\vec{R}$

Case - I : When $\theta = 0°$ i.e. $\vec{A}$ and $\vec{B}$ are in the same direction then from eqn (i), We have

$$R = \sqrt{A^2 + B^2 + 2AB\cos 0°}$$
$$= \sqrt{A^2 + B^2 + 2AB}$$
$$= \sqrt{(A+B)^2}$$
$$= A + B \text{ (maximum)}$$

Case - II : When $\theta = 180°$. i.e. $\vec{A}$ and $\vec{B}$ are in opposite direction then from eqn (i), we have

$$R = \sqrt{A^2 + B^2 + 2AB\cos 180°}$$
$$= \sqrt{A^2 + B^2 - 2AB}$$
$$= \sqrt{(A-B)^2}$$
$$= |A - B| \text{ (minimum)}$$

Date
Page

Case - III ÷ When $\theta = 90°$ i.e - $\vec{A}$ and $\vec{B}$ are perpendicular to each other then,

$$R = \sqrt{A^2 + B^2 + 2AB\cos 90°}$$
$$\therefore R = \sqrt{A^2 + B^2}$$

And,

From eqn (2)

$$\beta = \tan^{-1}\left(\frac{B\sin 90}{A + B\cos 90}\right)$$
$$= \tan^{-1}\left(\frac{B}{A}\right)$$

Q.n Which one set of forces is not possible?

(i) 1 N, 2N, 3N
(ii) 10 N, 10 N, 10 N
(iii) 5 N, 10 N, 20 N ✓
(iv) 5 N, 10 N, 12 N

☆ [$\because A - B \leq R \leq A + B$]

(3) Polygon law of vector addition

Statement ÷ It states that if a number of vectors acting at a point are represented both in magnitude and direction by the sides of polygon taken in same order then closing sides of the polygon taken in opposite order represents their resultant both in magnitude & direction.

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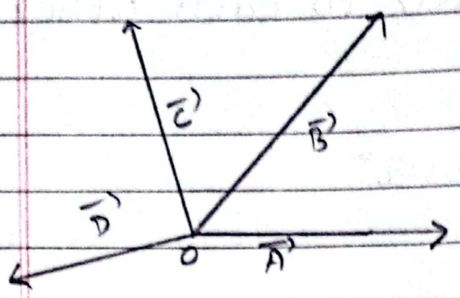
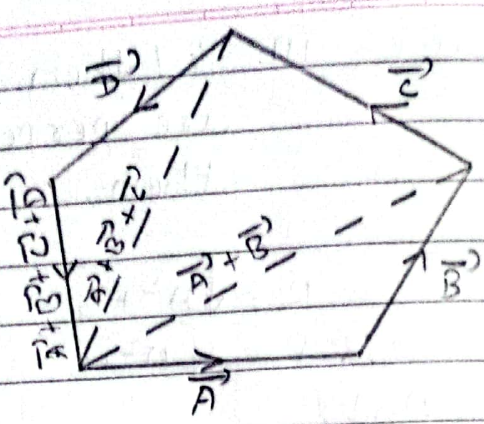
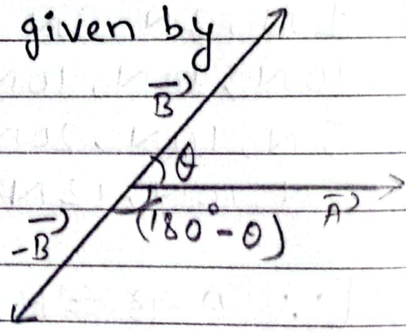




• Substraction of vectors

Consider two vectors $\vec{A}$ and $\vec{B}$ are inclined at an angle θ. Then substraction of $\vec{B}$ from $\vec{A}$ is given by $\vec{R} = \vec{A} - \vec{B}$

Then ~~R~~ magnitude of $\vec{R}$ is given by



$$R = \sqrt{A^2+B^2+2AB\cos(180-\theta)}$$

$$\therefore R = \sqrt{A^2+B^2-2AB\cos\theta}$$

$$[\because \cos(180-\theta) = -\cos\theta]$$

Qn. The magnitude of resultant of two equal vectors is equal to the magnitude of either vector. Find the angle between the two vectors.

→ Here,

According to question,

$|\vec{A}| = |\vec{B}| = |\vec{R}|$

i.e. $A = B = R = x$ (suppose)

Now,

or, $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

or, $R = \sqrt{r^2 + r^2 + 2r.r.\cos\theta}$

or, $r = \sqrt{r^2 + r^2 + 2r^2\cos\theta}$

Squaring on both sides,

$$\cancel{r^2} = \cancel{r^2} + r^2 + 2r^2\cos\theta$$

$$2r^2\cos\theta = -r^2$$

or, $\cos\theta = -\frac{\cancel{r^2}}{2\cancel{r^2}}$

or, $\cos\theta = -\frac{1}{2}$

or, $\theta = \cos^{-1}\left(-\frac{1}{2}\right)$

$$\theta = 120^\circ$$

$\therefore \theta = 120^\circ$ #.

Q.n. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} - \vec{B} = \vec{C}$ and $A - B = C$. Find the angle between them.

→ Soln,

Given,

$\vec{A} - \vec{B} = \vec{C}$

$A - B = C$

Then,

$C = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

or, $A - B = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

Squaring on both side,

$$(A-B)^2 = A^2 + B^2 - 2AB\cos\theta$$

or, $\cancel{A^2} - 2AB + \cancel{B^2} = \cancel{A^2} + \cancel{B^2} - 2AB\cos\theta$

or, $\frac{2AB}{2AB} = \cos\theta$

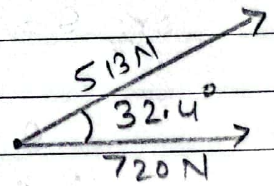
or, $\cos\theta = 1$

or, $\theta = \cos^{-1}(1)$

or, $\theta = \cancel{\theta}\ 0^\circ$

$\therefore \theta = 0^\circ$ #

① A rocket fires two engines simultaneously. one produces a thrust of 720N directly forward while the other gives a 513 N thrust at 32.4° above the forward direction. Find the magnitude and direction of the resultant force that ~~is~~ these engines exerts on the rocket.



→ Given,

$F_1 = 720\ N$

$F_2 = 513\ N$

$\theta = 32.4^\circ$

Now,

$R = \sqrt{(F_1)^2 + (F_2)^2 + 2.F_1.F_2.\cos\theta}$

or, $R = \sqrt{(720)^2 + (513)^2 + 2.720.513.\cos 32.4}$

or, $R = \sqrt{\cancel{233} \; 518400 + 263169 + 623721.9251}$

or, $R = \sqrt{1405290.925}$

$R = 1185.44$

$\therefore$ Resultant $(R) = 1185.44$

Again,

$$\theta = \tan^{-1}\left(\frac{\cancel{\sin}\ F_2 \sin\theta}{F_1 + F_2\cos\theta}\right)$$

$$\theta = \tan^{-1}\left(\frac{513\sin(32.4)}{720 + 513\cos 32.4}\right)$$

$$\theta = \tan^{-1}\left(\frac{274.87}{1153.14}\right)$$

$\theta = 13.4^\circ$

$\therefore \theta = 13.4^\circ$ #.

② Two forces 3N and 4N are acting perpendicular on ~~to~~ a body of mass 5 kg. Calculate the value of acceleration produced on the body.

→ Given,

$A = 3N$

$B = 4N$

$\theta = 90^\circ$

Mass (m) = 5 kg

We know that,

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

or, $R = \sqrt{3^2 + 4^2 + 2\times3\times4 \cdot \cos 90}$

or, $R = \sqrt{9 + 16 + 24\times0}$

or, $R = \sqrt{25}$

$\therefore R = 5$

$\therefore$ Resultant force = 5 N

Now,

$F = R = ma$

$5 = 5 \times a$

$a = \frac{5}{5}$

$\therefore a = 1 \ m/s^2$

$\therefore$ acceleration produced on the body $= 1 \ m/s^2$ #.

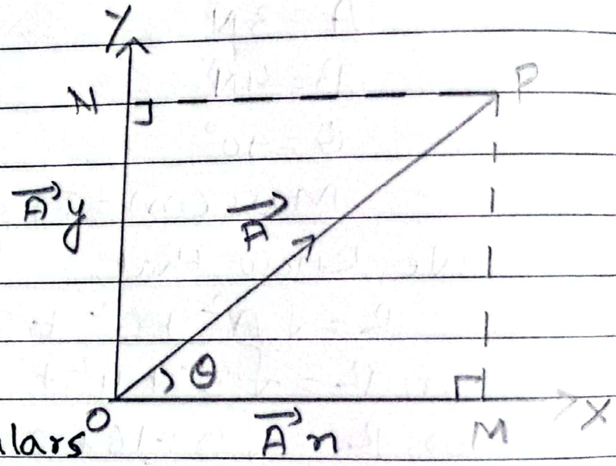
~~③ A disoriented~~

Resolution of a vector

↳ The phenomenon of splitting of a vector into two or more vectors is called resolution of a vector.

↳ It is opposite phenomena of addition of vectors.

Consider, a vector $\vec{A}$ is represented by $\vec{OP}$ in the XY-plane. let θ be the angle made by $\vec{A}$ with x-axis.



let us draw perpendiculars OM and ON on x-axis and y-axis respectively. Then,

$\vec{OM} = \vec{A}x$ and $\vec{ON} = \vec{A}y$ are the rectangular components of $\vec{A}$ along x-axis and y-axis respectively.

So, $\vec{A} = \vec{A}x + \vec{A}y$

$\therefore \vec{A} = Ax\vec{i} + Ay\vec{j} \longrightarrow ①$

From figure,

In ΔOPM,

$$\cos\theta = \frac{OM}{OP}$$

$$\text{or, } \cos\theta = \frac{Ax}{A}$$

$\therefore A_x = A\cos\theta \longrightarrow (2)$

And,

$$\sin\theta = \frac{PM}{OP}$$

$$\text{or, } \sin\theta = \frac{A_y}{A} \quad [\because PM = ON = A_y]$$

$$\therefore A_y = A\sin\theta \longrightarrow (3)$$

Squaring and adding equations (2) & (3) We get,

$$A_x^2 + A_y^2 = A^2\cos^2\theta + A^2\cdot\sin^2\theta$$

$$\text{or, } A_x^2 + A_y^2 = A^2(\cos^2\theta + \sin^2\theta)$$

$$\text{or, } A_x^2 + A_y^2 = A^2$$

$$\text{or, } A = \sqrt{A_x^2 + A_y^2}$$

$$\therefore A = \sqrt{A_x^2 + A_y^2} \longrightarrow (4)$$

This gives magnitude of $\vec{A}$.

Again,

Dividing eqn (3) by eqn (2) we get,

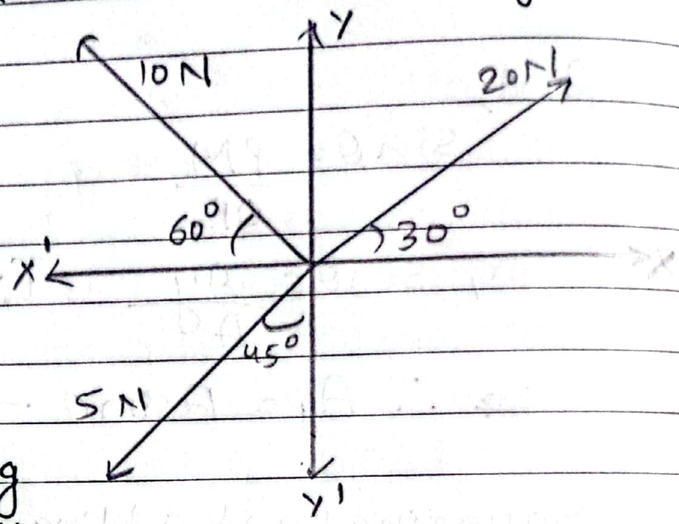
$$\frac{A_y}{A_x} = \frac{\cancel{A}\sin\theta}{\cancel{A}\cos\theta}$$

$$\text{or, } \frac{A_y}{A_x} = \tan\theta$$

$$\therefore \theta = \tan^{-1}\left(\frac{A_y}{A_x}\right) \longrightarrow (5)$$

This gives direction of $\vec{A}$

Q.n Find the Resultant force from the given figure.



→ Soln,

let, $F_1 = 20N$

$F_2 = 10N$

$F_3 = 5N$

Now,

Total force along x-axis is given by

$$F_x = F_{1x} + F_{2x} + F_{3x}$$

$$= F_1 \cos 30 + F_2 \cos 120 + F_3 \cos 225$$

$$= 20 \times \frac{\sqrt{3}}{2} + 10 \times -\frac{1}{2} + 5 \times -\frac{1}{\sqrt{2}}$$

$$= 10\sqrt{3} - 5 - \frac{5}{\sqrt{2}}$$

$$= 8.78 \ N$$

And,

Total force along y-axis is given by

$$F_y = F_{1y} + F_{2y} + F_{3y}$$

$$= F_1 \sin 30 + F_2 \sin 120 + F_3 \sin 225$$

$$= 20 \times \frac{1}{2} + 10 \times \frac{\sqrt{3}}{2} + 5 \times -\frac{1}{\sqrt{2}}$$

$$= 10 + 5\sqrt{3} - \frac{5}{\sqrt{2}}$$

$$= 15.12 \ N$$

Then,

Magnitude of resultant force is given by,

$$F = \sqrt{F_x^2 + F_y^2}$$

$$F = \sqrt{(8.78)^2 + (15.12)^2}$$

$$F = 17.48\ N$$

And,

Direction of Resultant force is given by,

$$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$$

$$= \tan^{-1}\left(\frac{15.12}{8.78}\right)$$

$$= 59.85^\circ$$

$$\therefore \theta = 59.85^\circ$$

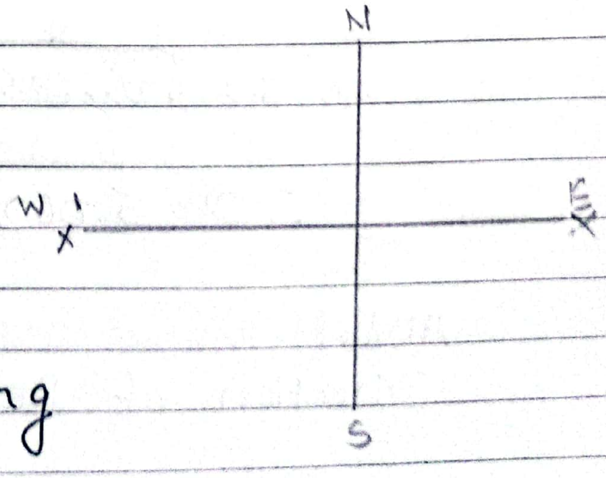
Q.n A disoriented physics professor drives 3.25 km north, then 4.75 km west and then 1.50 km south. Find the magnitude and direction of the resultant displacement by the method of resolution of Vector.

→ Soln,

$\vec{A} = 3.25$ km north

$\vec{B} = 4.75$ km west

$\vec{C} = 1.50$ km south



Now,

Total displacement along x-axis is

$$R_x = A_x + B_x + C_x$$

$= A\cos 90^\circ + B\cos 180^\circ + C\cos 270^\circ$

$\cancel{= 4.75 . \cos 90 +}$

$= 3.25\cos 90 + 4.75\cos 180 + 1.5\cos 270$

$= 0 + \left(-\frac{19}{4}\right) + 0$

$= -\frac{19}{4}$ km $= -4.75$ km

Total displacement along y-axis is

$R_y = A_y + B_y + C_y$

$= A\sin 90 + B\sin 180 + C\sin 270$

$= 3.25\sin 90 + 4.75\sin 180 + 1.5\sin 270$

$= 3.25 + 0 \cancel{+} - 1.5$

$= 1.75$ km

Then,

Magnitude of Resultant displacement is

$D = \sqrt{(R_x)^2 + (R_y)^2}$

or, $D = \sqrt{(-4.75)^2 + (1.75)^2}$

or, $D = \sqrt{22.5625 + 3.0625}$

or, $D = \sqrt{25.625}$

$\therefore D = 5.0621$ km.

And,

Direction of Resultant displacement is

$\theta = Tan^{-1}\left(\frac{R_y}{R_x}\right)$

or, $\theta = Tan^{-1}\left(\frac{1.75}{-4.75}\right)$

or, $\theta = Tan^{-1}\left(-\frac{7}{19}\right)$

or, $\theta = -20.22^\circ$

$\therefore \theta = 20.22^\circ$ north of west

• Multiplication of vectors

① Scalar Product or dot product of two vectors.

→ If the product of two vectors gives a scalar quantity then such operation is called scalar product.

The scalar products of two vectors $\vec{A}$ and $\vec{B}$ is given by

$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos\theta$

where, θ is the angle between $\vec{A}$ and $\vec{B}$.

The scalar product between two vectors obey commutative law

i.e. $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

Note: The angle between two vector is given by

$\theta = \cos^{-1}\left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}\right)$

2) Vector Product or cross product of two vectors

↳ If the product of two vectors gives a vector quantity then such operation is called vector product.

The vector products of two vectors $\vec{A}$ and $\vec{B}$ is given by

$$\vec{A} \times \vec{B} = |\vec{A}|\,|\vec{B}| \sin\theta \,\hat{n} = AB \sin\theta \,\hat{n}$$

where θ is the angle between $\vec{A}$ and $\vec{B}$ and $\hat{n}$ is the unit vector perpendicular to both $\vec{A}$ and $\vec{B}$.

The vector product between two vectors do not obey commutative law

i.e. $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$

but, $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$

Note: The magnitude of $\vec{A}$ and $\vec{B}$ is given by

$|\vec{A} \times \vec{B}| = AB \sin\theta$

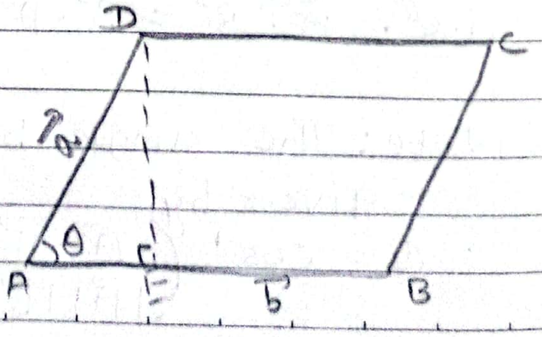
• Geometrical interpretation of vector product of two adjacent sides of a parallelogram gives the area of that parallelogram.

→ Proof

$$|\vec{a} \times \vec{b}| = |\vec{a}|\,|\vec{b}| \sin\theta$$

$$= AD \times AB \sin\theta$$

$$= AD \times AB \times \frac{DE}{AD}$$



$= AB \times DE$

$= \text{base} \times \text{height}$

$=$ Area of parallelogram ABCD #.

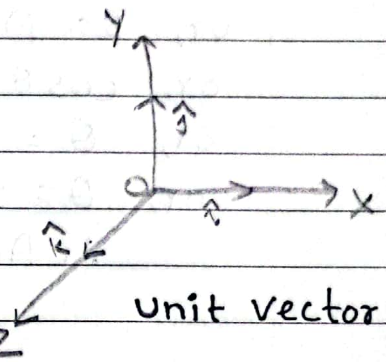
- Scalar and vector product of orthogonal unit vectors

→ ~~let~~ $\hat{i}.\hat{i} = |\hat{i}||\hat{i}| \cos 0° = 1 \times 1 \times 1 = 1$

Similarly,

$\hat{j}.\hat{j} = 1$

$\hat{k}.\hat{k} = 1$



b) $\hat{i}.\hat{j} = |\hat{i}||\hat{j}| \cos 90° = 1 \times 1 \times 0 = 0$

Similarly,

$\cancel{\hat{j}}$ $\hat{j}.\hat{k} = 0$

$\hat{k}.\hat{i} = 0$

Vector product

→ let $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vector along x-axis, y-axis and z-axis respectively.

a) $\hat{i} \times \hat{i} = |\hat{i}||\hat{i}| \sin 0° \hat{j} = 1 \times 1 \times 0 \times \hat{j} = 0$

Similarly, $\hat{j} \times \hat{j} = 0$

$\hat{k} \times \hat{k} = 0$

b) $\hat{i} \times \hat{j} = |\hat{j}||\hat{j}| \sin 90° \hat{k} = 1 \times 1 \times 1 \times \hat{k} = \hat{k}$

Similarly, $\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$

Since, $\vec{A}.\vec{B} = -(\vec{B}.\vec{A})$

so, $\hat{j} \times \hat{i} = -\hat{k}$

$\hat{k} \times \hat{j} = -\hat{i}$
$\hat{i} \times \hat{k} = -\hat{j}$

① If $\vec{A} \cdot \vec{B} = 0$, find the angle between $\vec{A}$ and $\vec{B}$.

→ Soln,

$\vec{A} \cdot \vec{B} = AB \cos\theta$

or, $0 = AB \cos\theta$

or, $\cos\theta = 0$

or, $\theta = \cos^{-1}(0)$

or, $\theta = 90°$

∴ Angle between $\vec{A}$ and $\vec{B}$ is $90°$.

2. $\vec{A}$ and $\vec{B}$ two non-zero vectors. If $|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$, what is the angle between $\vec{A}$ and $\vec{B}$?

→ Soln,

$|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$

or, $|A||B| \sin\theta = |A||B| \cos\theta$

or, $\sin\theta = \cos\theta$

or, $\dfrac{\sin\theta}{\cos\theta} = 1$

or, $\tan\theta = 1$

or, $\theta = \tan^{-1}(1)$

or, $\theta = 45°$

∴ Angle between $\vec{A}$ and $\vec{B}$ is $45°$.

3. The magnitude of two vectors are 3 and 4, and their dot product is 6. What is the angle between them.

→ Given,

$\vec{A} = 3$

$\vec{B} = 4$

$\vec{A} \cdot \vec{B} = 6$

$\theta = ?$

We know that,

$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}| \cdot \cos\theta$

or, $6 = 3 \times 4 \times \cos\theta$

or, $6 = 12 \times \cos\theta$

or, $\frac{1}{2} = \cos\theta$

or, $\theta = \cos^{-1}\left(\frac{1}{2}\right)$

or, $\theta = 60^\circ$

∴ Angle between two vector is 60°.